

CSE 311

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 2025
Program: B.Sc. in CSE
3rd Year 2nd Semester

Course Title: Data Communication and Computer Networks
Time: 1 hour

Credit Hour: 3.00

Course Code: CSE 311
Full Marks: 20

There are **Three** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- Show the key differences between LAN, MAN, and WAN considering the following: size (coverage), data transfer speed, application, and typical use cases. [3]
- Briefly describe the seven layers of the OSI model, including functions and at least two protocols of each layer. [4]
- Discuss the differences between Layer 2 and Layer 3 devices. Define each device and their role in communication network from **Figure 1** given below: [3]

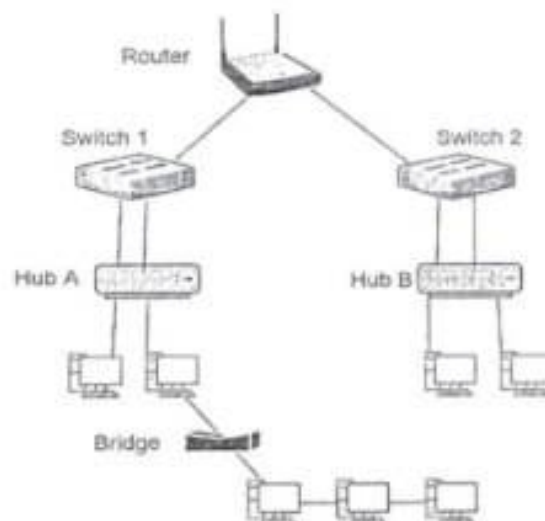


Figure 1. Basic Network devices by Layer

QUESTION 2 [10 MARKS]

- a. Why is line coding needed in digital communication? We have given a digital data bit as 010110101, you have to encode this data into the following line code scheme. [5]
- 1) Polar NRZ- L (Polar)
 - 2) Manchester (Dr. Thomas) (Polar)
 - 3) Alternate Mark inversion (AMI) (Bipolar)
- b. What is a subnet mask? Given IP Address: 172.16.50.200/19, find out the following: [5]
- a) Subnet mask.
 - b) Number of subnets.
 - c) Block size/host per subnet.
 - d) Number of valid hosts per subnet.
 - e) First and last valid host of each subnet.
 - f) Broadcast address.
 - g) Network address.

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Department of Computer Science and Engineering
Program: B.Sc. in CSE

Mid-Semester Examination Spring-2024 3rd year 1st Semester

Course Code: 303 Course Title: Data Communication Credit:3.0

Time: 1.00 Hour Full Mark: 20

There are Three Questions. Answer all of them. Part marks are shown in the margins.

1. a. Briefly describe the seven layers of the OSI (Open Systems Interconnection) model including each layer's primary function and protocols. **[4] [CO1]**
b. Show the key differences between Local Area Networks (LAN), Metropolitan Area Networks (MAN), and Wide Area Networks (WAN) in terms of size, coverage, data transfer speed, and typical use cases. Provide examples of where each type of network might be most appropriate. **[3] [CO1]**
2. a. Explain the different types of network topologies (e.g., bus, star, ring, mesh, hybrid etc.) and discuss their advantages and disadvantages. In your opinion, which topology is better for a large-scale network and why? **[3] [CO1]**
b. Investigate the key protocols in the IEEE 802.11 standard for Wireless Local Area Networks (WLANs), and compare them in terms of data rate, bandwidth, and frequency to demonstrate their role in promoting effective communication. **[2] [CO1]**
c. Consider all links in the network use TDM with 12 slots and have a data rate of 3.072 Mbps. Assume that host A takes 500 msec to establish an end-to-end circuit with host B before begin to transmit the file. If the file is 512 kilobytes, then how much time will it take to send the file from host A to host B? **[2] [CO2]**
3. a. Give a brief explanation of phase modulation (PM), frequency modulation (FM), and amplitude modulation (AM). Include the relevant figures or diagrams to demonstrate how the carrier wave is changed by each modulation method. **[3] [CO2]**
b. The equation of the amplitude for an analog wave is given by **[3] [CO2]**
$$S(t) = 20[1 + 0.8 \cos(2\pi \cdot 10^3 t)] \cos(4\pi \cdot 10^3 t)$$

Evaluate the following

 - I. The power of carrier signal
 - II. Modulation index of the modulated signal
 - III. Total sideband power.

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There are Two Questions. Answer all of them. Part marks are shown in the margins.

1. a. Describe which of the physical topology (BUS, STAR, RING or Mesh) will be applicable for the following scenario according to you and explain the reason: [5] [CO1]
- As a network administrator, you want to configure a high-speed LAN in your campus. Your network should be able to offer high performance, reliability, easy installation, expansion, basic security, and a single failure in PC connection does not affect the whole network.
- b. Describe the OSI model structure with proper figures, from the following point of view: [5] [CO1]
- In the OSI model, port address, logical address, and physical address are located in different layers. Data is also represented as different names such as segments, frames etc.
2. Construct the following digital signal diagrams, if you want to send 2 data packets 11111111 and 10101010 to your friend. Briefly **compare** among these line coding schemes and discuss which scheme would be better. Use examples where appropriate. [2* 5= 10] [CO2]
- Manchester
 - Differential Manchester
 - NRZ-I
 - AMI
 - Pseudoternary



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There are Three Questions. Answer all of them. Part marks are shown in the margins.

1. a.

[5] [CO1]



There is a network with a bandwidth of 1 Mbps. A message of size 1000 bytes has to be sent. A packet-switching technique is used. Each packet contains a header of 100 bytes. Out of the following, calculate the number of packets by which the message must be divided so that the total time taken is minimum-

- i. 1 packet
- ii. 10 packets
- iii. 20 packets

What network topologies should be considered for optimizing the connectivity and performance of the Computer Science and Engineering (CSE) department at the UAP campus? How can be the best solution identified?" [2] [CO1]

2. Construct the digital signal diagrams using the following line coding schemes if you want to send 2 data packets: 11111111 and 10101010 to your friend. Briefly compare these line coding schemes and discuss which scheme would be better. [6] [CO2]

- ~~i. Manchester~~
- ii. Differential Manchester
- ~~iii. NRZ-I~~

3. a. Specify the data communication model's architecture or its individual components. [2] [CO1]

b. Describe the OSI networking model, including an overview of each layer and the associated protocols. *at people seem to want Data Pack* [3] [CO1]

c. In wireless local area networks (WLANs), outline the major protocols in the IEEE 802.11 standard to the promote effective communication in terms of data rate, bandwidth, and frequency comparison? [2] [CO2]

*start-
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frequency
is like*

Department of Computer Science and Engineering
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Credit: 03

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Full Mark: 60

There are **Four** Questions. **Answer three questions including Q-1 and Q-2.**

1. ☒ a. Define the characteristics and criteria for a data communication system. [12]
- ☒ b. Draw the block diagram of a communication model and label each element of a communication system. What are the essential hardware requirements to establish a communication? [8]
2. ☒ a. Differentiate between: [8]
- i. Analog and digital signals
 - ii. Link and channel
 - iii. Protocol and Standard
 - iv. LAN and WAN
- ☒ b. What are the limitations of simplex communication? Compare half duplex and full duplex communication with merits and demerits. Mention at least three (3) examples of each. [12]
3. ☒ a. Define Network topology. [2]
- ☒ b. What are the basic topologies to setup a communication? Discuss the following issues for Bus, Mesh and Ring topologies: [18]
- i. Way of interconnection/ Diagram
 - ii. Transmission Mode (Simplex / Half duplex / Full Duplex)
 - iii. Type of link (point to point/ multi- point)
 - iv. No. of links
 - v. Supporting Device
 - vi. Robustness/ Security

OR

4. ☒ a. Define Network Architecture. [2]
- ☒ b. Draw TCP/IP Model and explain the functions of different layers to establish communication. [18]

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There are **Four** Questions. Answer three questions including Q-1 and Q-2.

1. a. Suppose, you are working as a network engineer in a Bank (even student ID)/ hospital (odd student ID). You know, that a network must be able to meet a certain number of criteria. Discuss the most important three of such criteria and elaborate how they are prioritized according to your working place. [10]

b. Suppose we consider a network of a single link, and we ignore transmission errors for now (the link is perfect). Suppose there are two hosts Ananta and Barsha and a file of 5000 bytes needs to be sent from Ananta to Barsha. The link bandwidth is 50Mbps, the link is a fiber optic link, and the distance between Ananta and Barsha is 25km. How long will it take to send the file from Ananta to Barsha? [10]

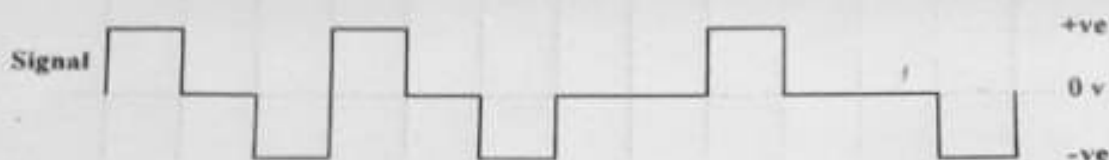
2. a. OSI model is a generic model that is based upon functionalities of each layer. TCP/IP model is a protocol-oriented standard. Briefly discuss the similarities and dissimilarities of these two entities with their layered approach in mind. [10]

b. Based on the following points, list the differences between Hub, Switch, and Router. [10]

Topic	Hub	Switch	Router
OSI Layer			
Function			
Data Transmission form			
No. of Ports			
Transmission type			
Device type			
Used in (LAN, MAN, WAN)			
Transmission mode			
Speed			

Address used for data transmission			
------------------------------------	--	--	--

3. a. Analyze the signals given below and determine the data bits for the following line coding schemes: [5*2=10]



An example of bipolar pseudoternary line coding



Manchester encoding (IEEE 802.3 version)

- b. In data communication, a line code is a pattern of voltage, current, or photons used to represent digital data transmitted down a communication channel or written to a storage medium. There are several line encoding schemes available such as: NRZ-L, NRZ-I, Manchester, multi-level encoding etc. Each scheme has a special significance. **Describe** the characteristics, significance of any two schemes with proper figures and examples. Also determine, which scheme better handles the DC component issue. You may use 101110-bit stream for your example. (Make your own assumptions for signal levels and declare it in your answer) [5*2=10]

OR

4. a. The 8B6T encoding scheme takes 8 bits of binary data and encodes them using 6 ternary signal elements. Recall that each ternary signal element can take on one of three values (positive, negative, and zero voltage). Table 1 is attached with this question (check the last page of this question paper) and is a portion of the 8B6T code table. Suppose the bit stream (in binary) 000100100101010001011000 is sent from sender to the receiver using 8B6T. Now answer the following: [2+2+6+10=20]
- What is the hexadecimal representation of this bit stream?
 - How many different bit patterns and signal pattern can be there?
 - Calculate the redundant signal elements and discuss their usage in synchronization and error control.

iv. Draw the digital signal diagram for 8B6T.

Data	Code	Data	Code	Data	Code	Data	Code
00	--+00-+	20	---+00	40	-00+0+	60	0++0-0
01	0---+0	21	+00+--	41	0-00++	61	+0+-00
02	0-+0-+	22	--0-++	42	0-0+0+	62	+0+0-0
03	0-++0-	23	+-0-++	43	0-0++0	63	+0+00-
04	--0+0-	24	+-0+00	44	-00++0	64	0++00-
05	+0---+0	25	-+0+00	45	00-0++	65	++0-00
06	+0-0-+	26	+00-00	46	00-+0+	66	++00-0
07	+0-+0-	27	++++--	47	00-++0	67	++000-
08	--+00+-	28	0++-0-	48	00+000	68	0++-+-
09	0-++-0	29	+0+0--	49	++-000	69	+0++--
0A	0-+0+-	2A	+0+-0-	4A	+-+000	6A	+0+-+-
0B	0-+-0+	2B	+0+--0	4B	--+000	6B	+0+--+
0C	--0-0+	2C	0++--0	4C	0+-000	6C	0+++++
0D	+0-+-0	2D	++00--	4D	+0-000	6D	++0+--
0E	+0-0+-	2E	++0-0-	4E	0-+000	6E	++0-+-
0F	+0--0+	2F	++0---0	4F	-0+000	6F	++0---+
10	0---+0+	30	+-00-+	50	+++0+	70	000++-
11	-0-0++	31	0+--+0	51	---0++	71	000+--
12	-0-+0+	32	0+-0-+	52	---+0+	72	000-++
13	-0-++0	33	0+-+0-	53	---++0	73	000+00
14	0---+0	34	+-0+0-	54	+++0+	74	000+0-
15	--00++	35	-0+-+0	55	--+0++	75	000+-0
16	--0+0+	36	-0+0-+	56	---+0+	76	000-0+
17	--0++0	37	-0++0-	57	---+0	77	000-+0
18	--0-+0	38	+-00+-	58	---0++	78	++++-0

Figure: 8B6T Partial Code Table

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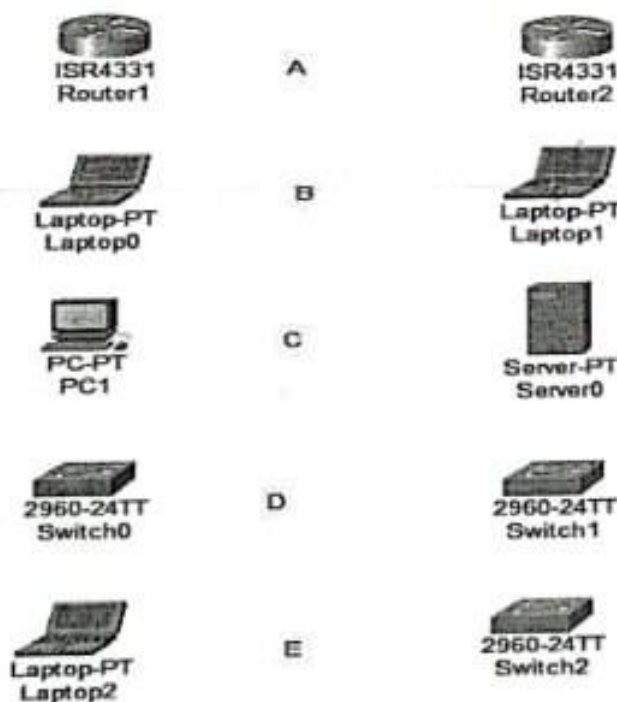
Credit: 3.00

Time: 1.00 Hour.

Full Mark: 60

There are **Four** Questions. Answer three questions including Q-1 and Q-2.

1. a. A data communication system has five components. **Define** them and describe their tasks with the proper figure. [10]
- b. You are creating an experimental setup with a few devices in your laboratory. **Analyze** the type of cable wiring you would consider between the following connections (A, B, C, D, E) so that proper data communication occurs. [10]



2. a. Suppose you want to measure the effectiveness of a data communication system that is running in your office. **Illustrate** the fundamental characteristics that needs to be considered during your measurement. [10]

- b. Based on the following points, list the differences between Hub, Switch, and Router. [10]

POINTS	Hub	Switch	Router
Layer of OSI model			
Main Function			
Data Transmission form			
No. of Ports (approximate)			
Transmission type			
Device intelligence type			
Used in (LAN, MAN, WAN)			
Transmission mode			
Speed			
Type of Address used for data transmission			

3. a. Apply your knowledge of IP addressing and find the error, if any, in the following addresses by applying existing rules of IPV4 format. You must support your answer with a proper explanation. If there is an error, rewrite it correctly, you may use any valid number in the wrong octet. If there is no error, justify whether it is a usable IP address or not. [2*5 =10]
- BA.23.14.67
 - 121.086.45.78
 - 75.45.31.50.314
 - 121.34.7.0
 - 255.255.255.255
- b. For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs). From your point of view identify these hardware, software along with their tasks. [10]

OR

4. a. There are four levels of addresses in the OSI model: IP address, MAC address, specific address, and port address. Each address is related to a particular layer of the OSI model. Describe the characteristics, significance of these addresses, and mention the corresponding layer name where they are dealt with proper figures and outlines. [20]

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There are **Four** Questions. **Answer three questions including Q-1 and Q-2.**

1. a.

[15]

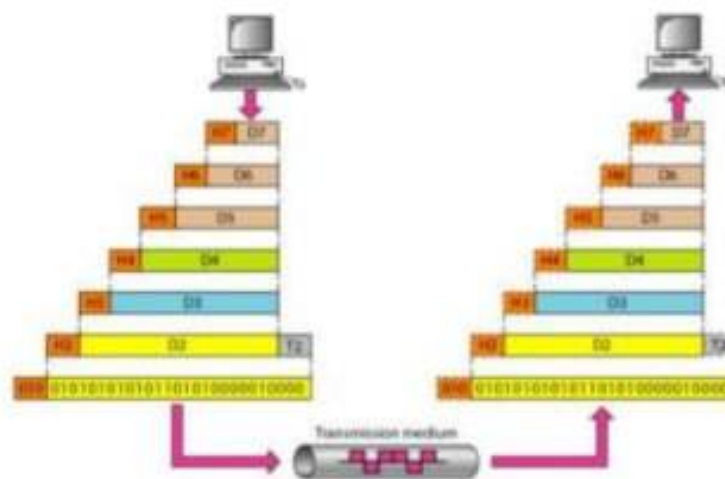


Figure 1 OSI layers

In OSI model, Data is represented by different formats and names starting from Application layer. It eventually turns into bits in the lowest layer and travels from sender to receiver and vice versa. Analyze, how Data encapsulates and decapsulates itself in different layers of OSI model starting from highest layer to the lowest layer with proper examples in each layer.

- b. Suppose you want to communicate with your friend, and you have a channel with 1000 bps bandwidth. You can choose either half-duplex data flow or full-duplex data flow for your communication. Which one will you choose? Evaluate the reason behind your choice. [5]
2. a. You are creating an experimental setup with few network devices in your laboratory. Analyze the type of cable wiring you would consider between the following connections (A, B, C, D, E) so that a proper data communication occurs. Also, find out error if any, in the provided figure. [10]

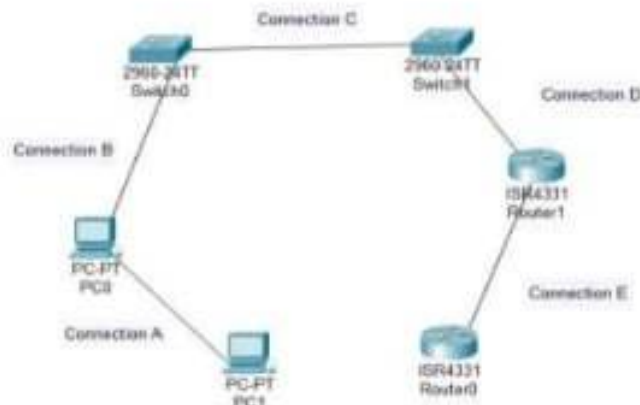


Figure 2 A simple Network topology

- b. Suppose you want to measure the effectiveness of a data communication system that is running in your office. Illustrate the fundamental characteristics that needs to be considered during your measurement. [10]
3. a. Determine the type of the following destination addresses. You must show the detailed calculation for each one. [3*2=6]
- 4C:30:10:21:10:1A
 - 49:20:1B:2E:08:EE
 - F2:FF:FF:FF:FF:FF
- b. The LAN market has seen several technologies such as Ethernet, Token Ring, Token Bus, FDDI, and ATM LAN. Some of these technologies survived for a while, but Ethernet is by far the dominant technology. In Standard Ethernet, the MAC sub layer governs the operation of the access method. It also frames data received from the upper layer and passes them to the physical layer. [10+4=14]
- Draw a MAC frame and elaborate its different fields. Justify the reasons behind minimum and maximum length of a MAC frame?
- OR**
4. a. There are two ranges: [10]
- 192.168.0.0 - 192.168.255.255
- 193.168.0.0 - 193.168.255.255
- Answer the following:
- From your point of view, is there any significance of these two IP ranges?
 - Mention key differences between these two ranges.

- iii) Who owns 193.168.0.0 IP address? (you can use internet for searching)
- b. "The physical addresses will change from hop to hop, but the logical addresses usually remain the same" -explain this statement with necessary facts and figures. **[10]**

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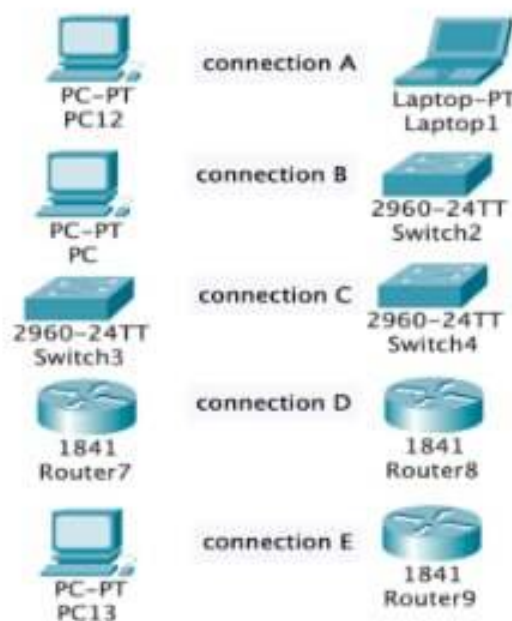
Time: 1.00 Hour.

Full Mark: 60

Instruction(s): Answer any three questions including 1 and 2.

1. a. Let's say you have several devices (PC, switch, router) to connect with a network to establish a successful data communication. [2+5+4]=11

- Explain the type of LAN technology you would use and why?
- Analyze the type of cable wiring you would consider for the following connections:



- If you want to connect an online, live video monitoring device (which will be placed far from main electricity power source) to the network, what would that device be and what changes you would need to bring in your cable wiring. Show the wiring diagram.
- b. Find the error, if any, in the following IPv4 addresses. You must support your answer with proper explanation. If there is an error, rewrite it correctly, you may use any valid number in the octet. If there is no error, justify whether it is a usable IP address or not. [1.5*6] = 9

- i) 111.056.45.78
- ii) 11100010.23.14.67
- iii) 0.0.0.0
- iv) 75.45.31.314
- v) 121.34.7.8.256
- vi) 255.255.255.255

2. a. Suppose you want to communicate with your friend, and you have a channel with 1000 bps bandwidth. You can choose either half-duplex data flow or full-duplex data flow for your communication. Which one will you choose? Evaluate the reason behind your choice. [8]
- b. Demonstrate the difference between bandwidth and throughput? Can throughput be greater than bandwidth? [6]
- c. "Like Data link layer, **Error control** and **Flow control** are also performed in Transport layer in end-to-end rather than one single link." – explain this statement. [6]

3. An address in a block is given as 192.168.17.X.
Subnet mask is : 255.255.255.192
Here,

$X = (\text{last digit of Your ID})^2 \bmod 6$,
For example, if your ID is 14101102 then,
 $X = 2^2 \bmod 6 = 4 \bmod 6 = 4$
So your IP address would be 192.168.17.4

- i) Find the number of addresses in the block size, number of subnets, and valid IP addresses in each subnet. [6+14=20]
- ii) Fill in the following table according to your calculations found above:

	Subnet 1	Subnet 2	Subnet N
Network address			
First Valid IP			
Last valid IP			
Broadcast address			

Or,

4. a. Suppose you are a data communication engineer, just started your career in a reputed company. You have been assigned a task to setup a network topology. You have two rooms to setup with two different topologies. One room has X computers connected with one type of topology and the other room has Y computers connected with a different topology.

[3+1+

Here, X = Summation of all digits of your ID
 Y = Last digit of your ID +2

6+4]
= [14]

For example, if your ID is 14101102, Then,
 $X = 1+4+1+0+1+1+0+2 = 10$
 $Y = 2 + 2 = 4$

Now answer the following:

- i) List all the topologies you might use in the above-mentioned scenario.
 - ii) Name any two topologies which you will actually use here.
 - iii) Calculate how many links/cables you will need for each of the topologies (according to your answer in "ii").
 - iv) If you were to choose between these two topologies (according to your answer in "ii") considering security and future growth which one will you prefer and why?
- b. Suppose you work in a network department of a hospital (**odd student ID**) / bank (**even student ID**). Your network must be able to meet a certain number of criteria. **[3+3 = 6]**
- i) What are those criteria?
 - ii) If you draw a pyramid figure of those criteria according to your work place, how it would look like? (You must draw the figure with proper outlines)

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Instruction(s): Answer any three questions including 1 and 2.

1. a. Suppose we have two topologies of computers in two rooms. One room has X computers connected with mesh topology and the other room has Y computers connected with star topology. [6+6=12]

Here, X is the last digit of your ID +1

Y is the second last digit of your ID +1

For example, if your ID is 14101109 then

$X = 9 + 1 = 10$

$Y = 0 + 1 = 1$

Calculate how many links you will need for each of the topologies.

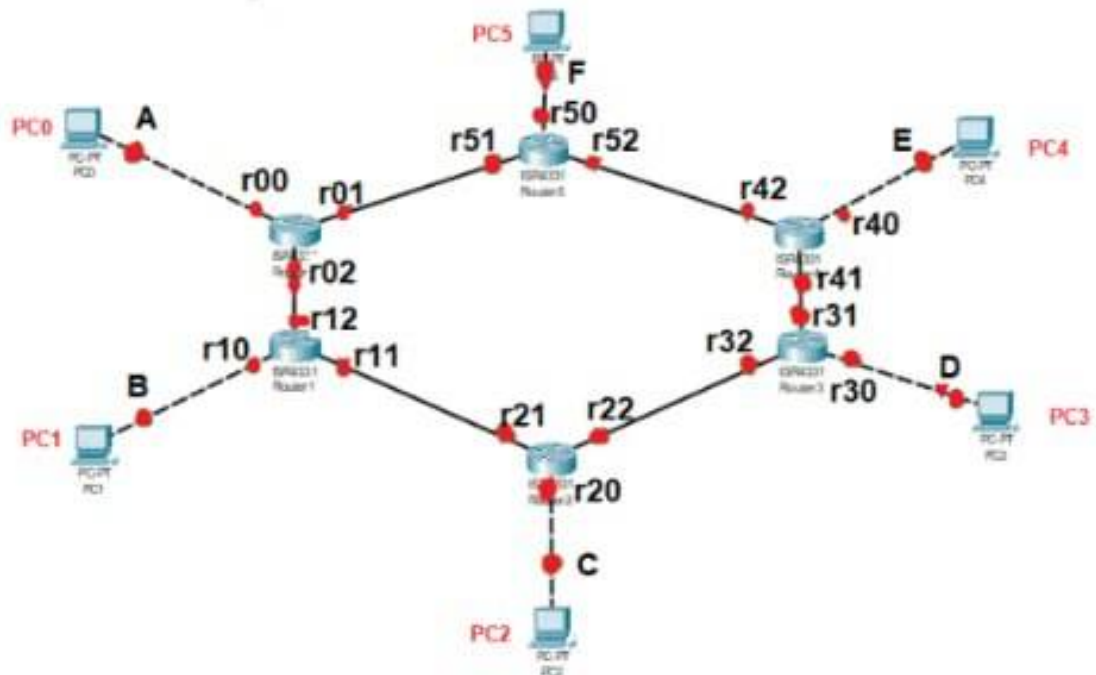
If you were to choose between these two topologies for a more secure network which one will you prefer and why?

- b. Suppose you want to communicate with your friend and you have a channel with 1000bps bandwidth. You can choose either half-duplex data flow or full-duplex data flow for your communication. [4x2=8]

Write one advantage and one disadvantage for each of these two data flows.

2. a. Suppose, six computers are connected through six routers in following topology.

[12]



You and your friend are sitting in two adjacent computers. You are in PCX and your friend is in PCY.

Here,

$$X = (\text{last digit of Your ID})^2 \bmod 6,$$

$$Y = (X+1) \bmod 6$$

For example, if your ID is 14101102 then,

$$X = 2^2 \bmod 6 = 4 \bmod 6 = 4$$

$$Y = (4+1) \bmod 6 = 5 \bmod 6 = 5$$

So you will be in PC4 and your friend will be in PC5

Here, each red points holds an IP address.

Points A, B, C, D, E and F correspond to 6 computers. Points r00, r01 and r02 correspond to 3 points of Router0.

PC0 is connected through Router0, PC1 is connected through Router1 and so on.

Suppose, you are sending a frame from Process1 (port no: 6000) in PCX. Your friend will receive the frame in Process2 (port no: 7000) in PCY.

Show the following parts of your frame along the travelling points.

Sender MAC	Receiver MAC	Sender IP	Receiver IP	Port no of sender process	Port no of receiver process	DATA (No need to change this component)	Trailer (No need to change this component)

**For simplicity no need to use any numeric value for IP and MAC addresses. If you want to write IP address of point A or r00, just write "IP of A" or "IP of r00".

**Same goes for the MAC addresses. For MAC address of point A, write "MAC of A". For MAC address of point r00 write "MAC of r00"

**If you follow the shortest path, there will be exactly six points along the travelling path.

- b. "Like Data link layer, **Error control** and **Flow control** are also performed in Transport layer in end-to-end rather than on single link." – explain this statement. [4+4=8]
3. a. Normally due to travelling some distance the power of a signal gets attenuated/loses its power. The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with -0.3 dB/km has a power of 2 mW, what is the power of the signal at X km? [6]

Here, X = last two digits of your ID

For example,

If your ID is 14101102 then X = 2

If your ID is 14101150 then X = 50

- b. "While the Shannon capacity gives us the upper limit; the Nyquist formula tells us how many signal levels we need"- Explain this statement. [8]
- c. Draw the followings: [6]
- Two signals with same amplitude and frequency but different phase
 - Two signals with same amplitude and phase but different frequency
 - Two signals with same phase and frequency but different amplitude

Or,

4. a. We have a channel with X MHz bandwidth. The SNR for this channel is $10 \cdot Y$ (Y multiplied by 10). What is the appropriate bit rate and signal level? [12]

Here, X is the last digit of your ID +1

Y is the second last digit of your ID +1

For example, if your ID is 14101109 then

$$X = 9 + 1 = 10$$

$$Y = 0 + 1 = 1$$

- b. What is the difference between bandwidth and throughput? Can throughput be greater than bandwidth? [8]

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QUESTION 1 [7 MARKS]

- a. Compare and contrast the advantages and disadvantages of using fiber optic cables versus copper cables for high-speed networks. Include factors like bandwidth, distance, cost, and security in your comparison. [3]
- b. Design a network for a large corporation with multiple branches across the city. Each branch must be connected to the central office. Which topology or combination of topologies would you recommend? Explain how this topology will ensure reliability and efficient communication across the branches. [4]

QUESTION 2 [6 MARKS]

- a. Discuss the differences between Layer 2 and Layer 3 devices. Define each device and explain their role in network communication from the Figure 1 given below: [7]

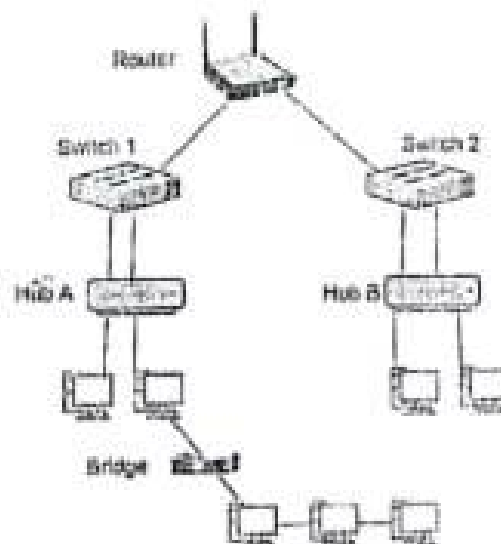


Figure 1. Basic Network devices by Layer

- b. Consider a network communication scenario where a user sends an email using an email client. Describe how the OSI layers interact during the process of sending and receiving this email, detailing the role of each layer in ensuring successful communication. [3]

QUESTION 3 [7 MARKS]

- a. A 10 MB file needs to be transmitted over a 1000 km optical fiber link in both circuit-switched and packet-switched networks. Consider the following: [4]
- Circuit-switched network:
 - Call setup time = 300 ms, Data rate = 100 Mbps, Propagation speed = 2×10^8 m/s
 - No queuing delay
 - Packet-switched network:
 - File is divided into 500-packet chunks, each 20 KB., Each packet has a processing delay of 2 ms per hop (4 hops), Data rate = 100 Mbps, Propagation speed = 2×10^8 m/s, No re transmission required

Which transmission method is faster?

- b. Given IP Address: 192.168.20.0/28, find out the followings [3]
- a) What is the subnet mask?
 - b) Find the number of Subnet
 - c) Block size/ Host per subnet
 - d) Number of valid hosts per subnet
 - e) 1st and last valid host of each subnet
 - f) What is the Broadcast address?
 - g) What is the Network address?

University of Asia Pacific
Department of Computer Science and Engineering
Program: B.Sc. in CSE

Mid-Semester Examination

Spring-2024

3rd year 2nd Semester

Course Code: CSE 319 Course Title: Computer Networks

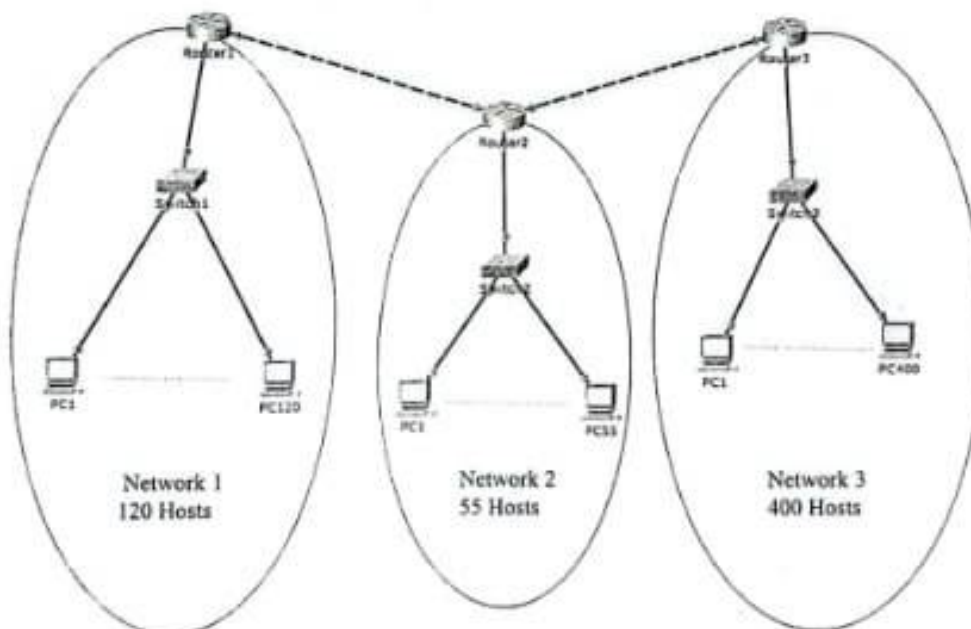
Credit: 3.0

Time: 1.00 Hour.

Full Mark: 20

There are **Two** questions. Answer all of them. Part marks are shown in the margins.

1. ✓ a. A software company has just bought an IP address of 190.168.0.0/22 and it needs to make subnets for the following network topology: [5] [CO3]



Calculate the network address, subnet mask, valid host size, and broadcast address of each subnet. You need to make the subnets keeping in mind that there should be very minimal wastage of host IP addresses.

- b. In a packet-switching network, packets are routed from source to destination along a single path having two intermediate nodes. The bandwidth of the network is 500 KBps. If the message size is 500 bytes and each packet contains a header of 50 bytes, in how many packets the message must be divided so that the total time taken is minimum? You can make 1 packet, 5 packets, or 10 packets. [3] [CO1]
- ✓ c. Write down the differences between circuit switching and packet (datagram) switching. [2] [CO1]

2. You are a network administrator in charge of managing and optimizing a university's network, including the configuration of various protocols, port numbers, and DNS services. Your role involves ensuring smooth communication, data transfer, and user access to university resources, including the website www.uap-bd.edu. Now answer the following questions, keeping in mind that your answers should be clear and well-structured, providing examples where applicable. Use diagrams if necessary to illustrate complex concepts, especially in the DNS resolution process.
- a. Briefly discuss the usage of any four well-known application layer protocols, their port numbers, and associated transport layer protocols. [2] [CO1]
 - b. Explain the working principle of a DNS server when a user searches for the website www.uap-bd.edu. Outline the steps involved in resolving this domain name into an IP address, including any interactions with root DNS servers, TLD servers, and authoritative DNS servers. Be sure to explain the significance of each step in the process. [5] [CO1]
 - c. Compare and contrast the roles and functionalities of SMTP, POP, and IMAP in email communication. Analyze how each protocol handles email transmission, retrieval, and storage, and evaluate their advantages and limitations in different use cases. [3] [CO1]

University of Asia Pacific
Department of Computer Science and Engineering
Mid-Semester Examination Fall-2022
Program: B.Sc. in CSE

Course Title: Computer Networks

Course No. CSE 319

Credit: 3.00

Time: 1.00 Hour.

Full Mark: 60

There are **Four** Questions. Answer three questions including Q-1 and Q-2.

1. Whenever we send the data from one node to another in a computer network, the data is encapsulated at the sender's side, while it is de-encapsulated at the receiver's end. Actually, the encapsulation of data at various layers of the implementing model (OSI or TCP/IP) adds various functionalities and features to the data transmission. **Illustrate** several steps of encapsulation and de-encapsulation with the proper figure. [15] CO1

2. The distance from Teknaf to Tetulia is approximately 1000 Km. If the propagation speed is 2.4×10^8 m/s in the cable, estimate the propagation time and the transmission time for a 2.5kbyte message (an e-mail) if the bandwidth of the network is 1 Gbps. [5] CO1



The above network is having a bandwidth of 1 MBps. A message of size 5000 bytes has to be sent. Each packet contains a header of 60 bytes. Out of the following, identify in how many packets the message must be divided so that the total time taken is minimum if the packet switching technique is used:

- 1 packet
3 packets
5 packets
10 packets
3. A routing table contains the entry of a subnet mask for every network. If there are many small networks then the size of the routing table increases. When the router has a big routing table then it takes a lot of time for the router to process the routing table. Supernetting is used to reduce the size of the IP routing table to improve network routing efficiency. Now consider the following IP addresses and determine whether they can be aggregated or not. [20] CO2
- 201.1.0.0/24
201.1.1.0/24

201.1.2.0/24

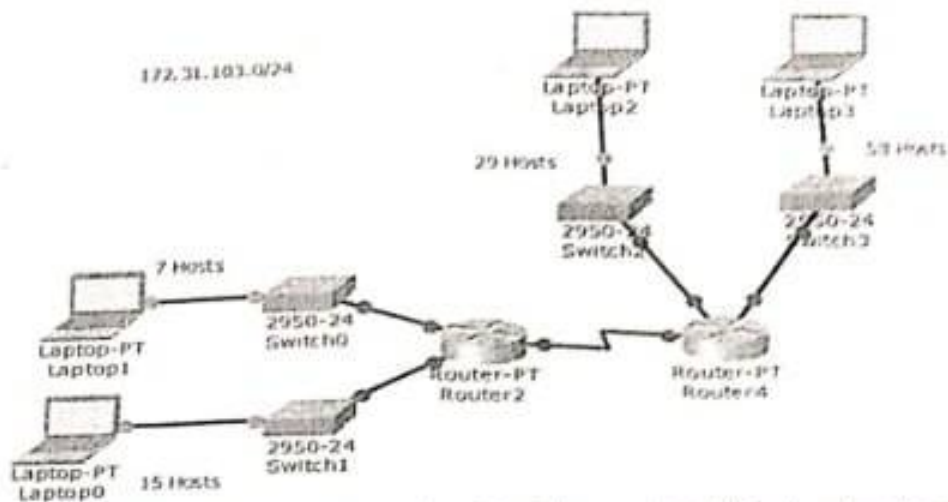
201.1.3.0/24

You need to mention step-by-step procedures while solving this question.

OR

4.

[20] CO2



Consider the above topology diagram of an office. There are in total four departments and they have different Host requirements (indicated beside the links). You should also consider the area between the two routers. Examine the above network diagram and derive the solution according to the following table:

	Department 1	Department N	Router to router
Network Address			
First usable host			
Last usable host			
Broadcast address			
Subnet mask			

University of Asia Pacific

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Mid-Semester Examination

Fall-2023

3rd year 2nd Semester

Course Code: CSE 319

Course Title: Computer Networks

Credit: 3.0

Time: 1.00 Hour.

Full Mark: 20

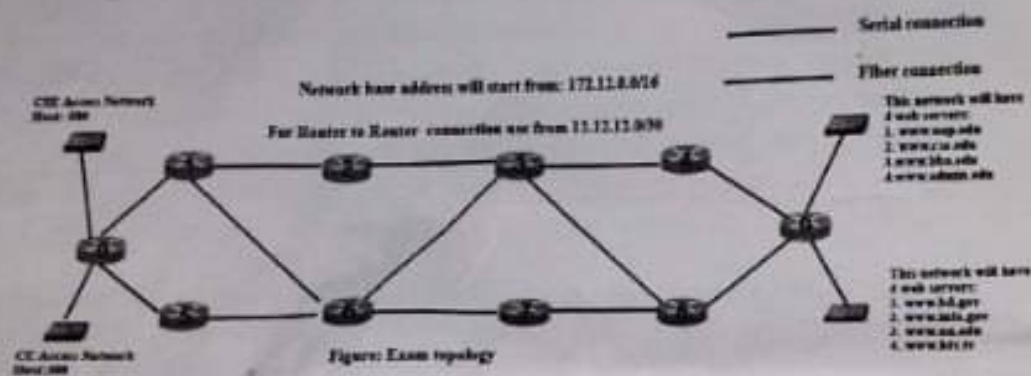
There are Two Questions. Answer all of them. Part marks are shown in the margins.

1. Simple Mail Transfer Protocol (SMTP) is an application layer protocol that is used to transmit electronic mail. Besides the facilities provided by SMTP, we may still face some problems.
 - a. Explain the limitations of SMTP. [2] [CO1]
 - b. Describe how these can be solved by Multipurpose Internet Mail Extensions (MIME). [4] [CO1]
 - c. Briefly interpret the working process and Base64 encoding of MIME. [4] [CO1]
2. In pipelined protocols, the sender is allowed to transmit multiple packets before having to wait for acknowledgements. So, the utilization of the sender is essentially multiplied.
 - a. Demonstrate how pipelining can improve the performance of RDT 3.0. [4] [CO2]
 - b. In Go-Back-3, if every 5th packet being transmitted is lost and we have to send 10 packets, then identify the required number of transmissions with proper figures. [6] [CO2]

distribute the address among the 32 ISP's of Bangladesh with the same number of host capabilities. Design the sub-network and list those addresses.

Or

- Q.4 IT department of the University of Asia Pacific (UAP) plans to increase the network performance and divide the whole network into smaller sub-networks for better management as shown in the following Figure. (20)



As a network engineer, your job is to design the UAP network with each sub-network's network address, sub-net mask, usable host range, and broadcast addresses from the above requirements.

University of Asia Pacific

Department of Computer Science & Engineering

Mid-Semester Examination Spring 2021

Program: B.Sc. Engineering (3rd Year/2nd Semester)

Course Code: CSE 319

Course Title: Computer Networks

Credit: 3.00

Time: 1.00 Hours

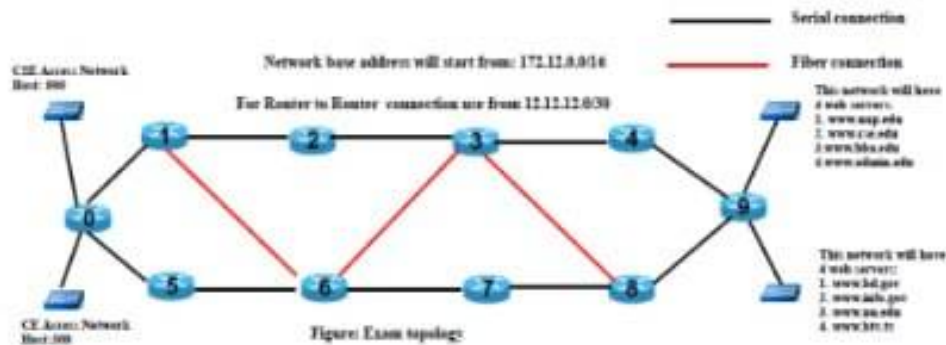
Full Marks: 60

Instructions: There are Four Questions. Answer three questions including Q-1 and Q-2. All questions carry equal marks.

- Q. 1 a) Suppose you want to send a large file to your friend B. The path from you to B has three links, of rates $R_1 = 500$ kbps, $R_2 = n \times 200$ kbps, and $R_3 = 800$ kbps, where $n = \text{your birth month}$. Let assume there is no other traffic in the network. Identify the throughput for the file transfer from you to your friend B? (5)
- b) TCP/IP protocol stack ensures the end to end communication where each layer performs its tasks. Categorize and describe the functionality of TCP/IP layers with their corresponding protocols. (15)
- Q. 2 a) Let two network applications communicate over the unreliable network by ensuring that both parties will get the data reliability service from the underlying layer. Here data reliability has the following criterion:
(i) detect flip bits
(ii) recover from packet loss
(ii) ensures data security
For the above requirements, design a robust protocol with its finite state machine (FSM) specification. (10)
- b) IT department of the University of Asia Pacific (UAP) received some complaints from the student that they cannot get enough browsing speed while surfing. In addition, the IT department wants to know the most visited site from the campus.
Propose a unified solution and evaluate your proposal to justify your recommendation. (10)
- Q. 3 a) List the subnetwork address, subnet mask, and broadcast address from the following IP addresses: (9)
(i) 162.12.5.6/a
(ii) 10.10.10.10/b
(iii) 192.1468.54.250/c
Here, $a = 15 + \text{Your birth month}$
 $b = 17 + \text{Your birth month}$

$$c = 18 + \text{Your birth month}$$

- b) Let host A establishes a connection with host B for the application layer requirement since the application needs the reliability feature for data transmission. Describe the process for reliable data transmission and explain why it requires three-way handshaking to establish the connection. (8)
- c) Identify how many subnetworks are in the following network topology? (3)



Or

- Q.4 IT department of the University of Asia Pacific (UAP) plans to increase the network performance and divide the whole network into smaller subnetworks for better management. The IT department identified that UAP requires seven subnetworks, namely CSE, CE, EEE, Phy, Eng, Law, and Admin. Table 1 describes the host requirements for each subnetwork. (20)

Table 1: Host requirements	
Department	Number of Hosts requirements
CSE	60 <i>n</i>
CE	11 <i>n</i>
EEE	13 <i>n</i>
Phy	51 <i>n</i>
Eng	40 <i>n</i>
Law	84 <i>n</i>
Admin	12 <i>n</i>

As a network engineer, your job is to design the UAP network with each subnetwork's network address, subnet mask, usable host range, and broadcast addresses from the above requirements. **You have to keep in mind that with each subnetwork multiply your birth month to get the actual host numbers.**

To calculate the subnetwork packages, choose the base network address from Class A private address if your student id is odd otherwise choose Class B private address.

University of Asia Pacific

Department of Computer Science & Engineering

Mid-Semester Examination Fall 2020

Program: B.Sc. Engineering (3rd Year/2nd Semester)

Course Code: CSE 319

Course Title: Computer Networks

Credit: 3.00

Time: 1.00 Hours

Full Marks: 60

Instructions: There are Four Questions. Answer three questions including Q-1 and Q-2. All questions carry equal marks.

Q. 1 XYZ ISP corporation shares n Gbps access link to provide Internet connectivity for its customers. They have a standard internet package, which is 10 Mbps@500 BDT per month. In addition, the administrator of the XYZ ISP observes that 80% of the time access link is idle.

$n = \text{Your birthday.}$

- a) When circuit switching is used, how many users can be supported, and how much money can ISP earn? (5)
- b) Can the XYZ ISP provide more connections based on the above scenario that you have found in question number a? How? Let the ISP administrator identifies that 250% more users can be connected than the circuit switching techniques with the standard internet package. Calculate the probability that at any given time, all active users are transmitting simultaneously. (5+10)

Q. 2 Mr. X is an independent consultant who provides consultancy services for subnetwork designing. Janata Bank Bangladesh Limited (JBBL) is searching for a resource person who can provide the consultancy service to increase its network reliability. JBBL needs to manage its branches by their geographic location that they can effectively monitor the network performance. (20)

Mr. X has identified that JBBL could break its network into eight subnetworks based on eight-divisions of Bangladesh. However, each division has its requirements for the hosts. From the analysis, Mr. X has discovered that the Dhaka division requires the maximum number of hosts. Table 1 describes the Union Councils in each division, and for one union council, at least i hosts are required for that particular subnetwork.

Table 1: Bangladesh Divisions' data

Division	Number of Districts	Number of Union Councils
Barisal	6	333
Chittagong Division	11	949
Dhaka Division	13	1,248
Khulna Division	10	270
Mymensingh Division	4	350
Rajshahi Division	8	558
Rangpur Division	8	536
Sylhet Division	4	334

Based on the above and the following requirements, design the network as Mr. X will design for the JBBL.

Let the number of union council in each division is u ; then

$i = u \times \text{your birth month}$ (for example: For Barisal Division $u = 333$ and your birth month is April; then $i = 333 \times 4 = 1332$)

To calculate the subnetwork packages, choose the base network address from Class A if your student id is odd otherwise choose Class B private addresses.

- Q. 3 a) Consider a TCP connection between Host A and Host B. Suppose that the TCP segments traveling from Host A to Host B have source port number is "your id" and destination port number is "your birth day". Identify the source and destination port numbers for the segments traveling from Host B to Host A. (5)
- b) Suppose a process in Host C has a UDP socket with port number 6789. Suppose both Host A and Host B each send a UDP segment to Host C with destination port number 6789. Will both of these segments be directed to the same socket at Host C? If so, how will the process at Host C discover that these two segments originated from two different hosts? (10)

- c) Explain why an application developer might choose to run an application over UDP rather than TCP. (5)

Or

- Q. 4 A network bottleneck refers to a discrete condition in which data flow is limited by computer or network resources. The flow of data is controlled according to the bandwidth of various system resources. If the system working on a network delivers a higher volume of data than what is supported by the network's existing capacity, then a network bottleneck will occur. (20)

In web access, we may face a bottleneck between the web server and the client. Figure 1 illustrates a bottleneck scenario between an institutional network and a web server. What is your proposal to improve the bottleneck scenario for the institutional network? Explain with evidence and example.

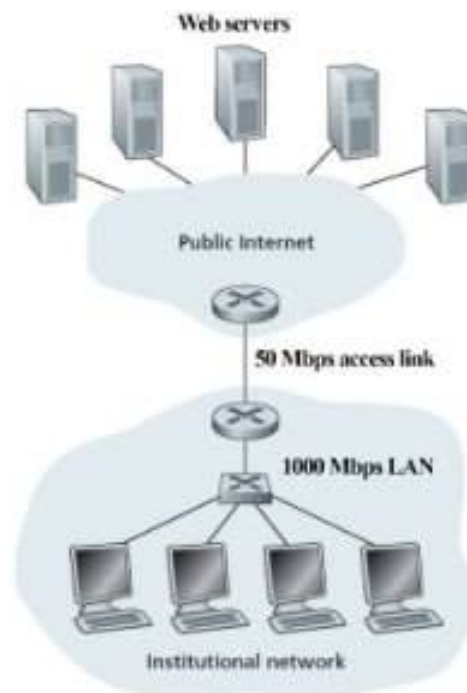


Figure 1: Bottleneck scenario

University of Asia Pacific

Department of Computer Science & Engineering

Mid-Semester Examination Spring 2020

Program: B.Sc. Engineering (3rd Year/2nd Semester)

Course Code: CSE 319

Course Title: Computer Networks

Credit: 3.00

Time: 1.00 Hours

Full Marks: 60

Instructions: There are Four Questions. Answer three questions including Q-1 and Q-2. All questions carry equal marks.

All questions are based on your registration ID. Therefore, to answer the question, you have to use your registration ID and other necessary information.

- Q.1 During this lockdown, we have to stay at home and use the internet to communicate with each other. We are living in different regions of the country, and the Internet Service Provider (ISP) of each area provides different speeds to its customers. Suppose you are Host A, and your friend is Host B, and between A and B, there are three communication links, namely, R_1 , R_2 , and R_3 . The speed of the connection depends on your registration ID, and you have to calculate the link speed based on the following equations:

$$\begin{aligned}R_1 &= (\text{your Id}) \bmod 49 \text{ mbps} \\R_2 &= (R_1 + \text{your birth day}) \text{ mbps} \\R_3 &= \frac{R_1 + R_2}{2.0} \text{ mbps}\end{aligned}$$

For example, your id is 14201041 and birthday is 12 April, then $R_1 = (14201041) \bmod 49 = 8 \text{ Mbps}$; $R_2 = 8 + 12 = 20 \text{ Mbps}$; $R_3 = (8 + 20) / 2 = 14 \text{ Mbps}$

Based on the above scenario, answer the following questions:

- a) Assume that there is no other traffic in the network, what is the throughput between you and your friend for file transfer? (8)
- b) Suppose you want to send a zip file, which is the double size in Gigabit (Gb) of your birth year (i.e., for the birth year 2001, the file size will be 4002 Gb) to your friend. How long will it take to transfer from you to your friend? (12)
- Q.2 a) You and your friend want to transfer data, and both are decided on the following data transmission properties which should be incorporated in the transport layer service: (10)
- no bit errors; no packet loss; encryption at the sender and decryption at the receiver.

Now design the Finite State Machine (FSM) for you and for your friend.

- b) Due to coronavirus and lockdown, most people stay at home and use the internet for necessary communications. However, many people are watching movies and videos, which have increased the traffic on the network. In this situation, what will you propose to improve the performance of the network? Explain with evidence. (10)

- Q. 3 a) Suppose IP address of your computer is as your registration ID (*exception for id number 17201100, you will use 17201122*), i.e., 14.20.10.41 where the network mask is based on the following formula
Netmask = 16+your birth month; for example, your birth month is April, and the netmask will be 16+4=20. Therefore, the address will be 14.20.10.41/20. Now do the following: (5+5)
- Find the broadcast address of the network
 - The address range of the network
- b) In data transmission, it is necessary to check the data at the receiving end that it has any error or not. Suppose we want to send your registration id (14201041), and it is breaking into two parts A=1420; B=1041. Now we will convert it into binary based on the following rules: (10)
- The number between 0 to 4 will be binary 1
The number between 5 to 9 will be binary 0
Therefore, A and B will be A=1111; B= 1111
However, now you have to send the data and do the process that at the receiving side we can detect if there is any error or not.

Or

- Q. 4 You are appointed as the network designer of the University of Asia Pacific. From your experience, you have identified that for better performance of the network, it should break in small LANs based on functionality. Therefore, you have planned to set the individual LAN for each department, and it is described in Table 1. To calculate each department's subnetwork address packages, choose the base network address from your registration id's first six numbers (considering your id is 14201010) for the first three octets as like 14.20.10, and for the last octet, choose 0 with the subnet mask 23, i.e., 14.20.10.0/23. (20)

Table 1: Requirements of each LAN

Department	Number of Hosts
CSE	100
CSE	150
CE	130
Architecture	70
Pharmacy	32
Admin	15

CSE 315
University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 25
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Microprocessors and Microcontrollers
Time: 1 hour. Credit Hour: 3.00

Course Code: CSE 315
Full Marks: 20

There are Two Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Draw the physical memory system of a microprocessor with a 24-bit data bus and 20-bit address bus, and calculate its total memory size. [4]
- b. 1. Classify the addressing mode of each instruction and 2. Calculate the physical address where applicable (DS = 3000H). [2+4]
- (i) MOV [4333H], AX
 - (ii) MOV [BX+4H], AX
 - (iii) MOV AL, 2564H
 - (iv) MOV CH, [BX]
 - (v) MOV AX, BX
 - (vi) MOV DX, [SI]

QUESTION 2 [10 MARKS]

You are given the following pseudocode:

Consider a = AL, b = BL, c = CL

Start

a ← 3AH

b ← 7CH

c ← a - b

c ← c + 1

a ← NOT a

$b \leftarrow -b$

swap(a, b)

$c \leftarrow c \oplus b$

$c \leftarrow c \vee b$

$c \leftarrow c \wedge a$

End

- a. Modify the given pseudocode into an equivalent Assembly program. [5]
- b. For each instruction in your assembly program, determine which flags (CF, ZF, SF, OF, PF, AF) are affected and show their status (0 or 1) after execution. [5]
Assume all flags are initially cleared (0) before execution.

University of Asia Pacific

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Mid-Semester Examination

Spring-2024

3rd year 1st Semester

Course Code: CSE 311 Course Title: Microprocessors and Assembly Language

Credit: 03

Time: 1.00 Hour

Total Marks: 20

There are **Three** Questions. Answer all of them. Part marks are shown in the margins.

1. a. What are the mentionable characteristics of a general microprocessor? If a microprocessor is 64-bit, then mention the size of
i. Register
ii. Data bus
iii. ALU [3] CO1
b. Draw the block diagram of 8086 architecture. [3] CO1
2. a. Suppose AL= 42H and BL= (last two digits of your ID) Hex, find out the situation of status flags for the following instructions: [3] CO1
i. MOV AL, BL
ii. ADD AL, BL
iii. DEC AL
b. Explain the role of SP during stack operation, if SP= 0100H, then show with a diagram the change of SP during PUSH and POP operation. [3] CO1
3. a. Find out legal or illegal statements from the followings and mention the reason in one line: [3] CO2
i. MOV AX, BL
ii. MOV AL, [BX]
iii. MUL A, B
iv. ADD X, Y
v. SUB BL, B
vi. INC 5
b. Write an assembly program that calculates the equation: [3] CO2
 $R = 2x - 3y$, where $x = 4$ and $y = 2$
c. Write instructions for the following: [2] CO2
i. Taking an input from keyboard
ii. Display the first character of your name on monitor
iii. Execute newline with carriage return
iv. Program exit to DOS

University of Asia Pacific
Department of Computer Science and Engineering
Program: B.Sc. in CSE

Mid-Semester Examination

Spring-2023

3rd year 1st Semester

Course Code: CSE 311 Course Title: Microprocessors and Assembly Language

Credit: 03

Time: 1.00 Hour

Total Marks: 20

There are **Three** Questions. Answer all of them. Part marks are shown in the margins.

1. ☒ a. Write down the basic features of 8086 microprocessor. What are the differences between port and register? [3] CO1
- ☒ b. Draw the block diagram of 8086 architecture. [3] CO1
2. ☒ a. Explain the followings for a microprocessor: [3] CO1
 - ☒ i. 'A microprocessor is 4-bit' - What features you can conclude from this statement?
 - ☒ ii. 'A microprocessor has 64KB memory (RAM)' - What is the length of address bus? [Compute from the memory capacity].
- ☒ b. Explain using necessary diagram that stack operated in LIFO manner using SP and code segment operated in FIFO using IP. Mention the procedure to calculate physical address. [3] CO1
3. ☒ a. Find out legal or illegal statements from the followings and mention the reason in one line: [3] CO2
 - i. MOV AX, BL
 - ii. MOV AL, [BX]
 - iii. MUL A, B
 - iv. ADD X, Y
 - v. IN AL, [DX] ✗
 - vi. INC 5
- ☒ b. Write an assembly program to find out the area of a rectangle. Consider the variables in byte form. [3] CO2
- ☒ c. List out the utility of MODEL and PROC. [2] CO2

University of Asia Pacific

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Mid-Semester Examination

Fall-2023

3rd year 1st Semester

Course Code: CSE 311 Course Title: Microprocessors and Assembly Language

Credit: 03

Time: 1.00 Hour

Total Marks: 20

There are Three Questions. Answer all of them. Part marks are shown in the margins.

1. a. Draw block diagram of a Basic Computer System. Discuss Life Cycle of a Microprocessor. [4] CO1
- b. Differentiate between the followings: [3] CO1
 - i. Port and Register
 - ii. Stack and Queue
 - iii. AX and DX
2. a. Suppose AL = 0FFH and BL = (last two digits of your ID) Hex, find out the situation of status flags for the following instructions: [3] CO1
 - i. MOV AL, BL
 - ii. ADD AL, BL
 - iii. NEG AL
- b. Write down the functions of BIU and mention the elements [2] CO1
- fetch, decode, prefetch, write
3. a. Write the assembly code for the following actions: [6] CO2
 - i. Input the first letter of your name
 - ii. Display the letter in the next line
 - iii. Display your name as a string in the next line with a beep sound
- b. Write an assembly program to swap two 8-bit numbers using a third register. [2] CO2

University of Asia Pacific

Department of Computer Science and Engineering

Mid-Semester Examination Fall-2022

Program: B.Sc. in CSE

Course Title: Microprocessors and Assembly Language

Course No. CSE 311

Credit: 03

Time: 1.00 Hour.

Full Mark: 60

There are Four Questions. Answer three questions including Q-1 and Q-2.

1. a. Draw block diagram of a microcomputing system. What are the functional units of a general microprocessor? Define those. [8]
- b. A microprocessor has a memory capacity (RAM) of 32KB (32X8). Construct 32KB capacity by using four 8KB (8 X8) chips. Show the necessary connections and address mapping. [12]
2. a. List out the functions of the followings: [10]
 - i. Instruction Pointer
 - ii. Queue
 - iii. Stack Pointer
- b. Explain the functions of AX and DX register during multiplication and division operations with suitable example. [10]

[Consider both byte and word operation]
3. a. Write appropriately the meaning of the following statements: [10]
 - i. .MODEL SMALL
 - ii. .STACK 100H
 - iii. MAIN ENDP
 - iv. A DW ?
 - v. INT 21h
- b. Write an assembly code to input a character and display it in the next line. [10]

OR

4. a. Define the six conditional flags. [10]
- b. Suppose AL= FFH and BL= AL= (last two digits of your registration number) H. Then show the status of six conditional flags after execution of ADD AL, BL instruction. [10]

[Mention the reasons]

University of Asia Pacific
Department of Computer Science and Engineering
Mid-Semester Examination Spring-2022
Program: B.Sc. in CSE

Course Title: CSE 311

Course No. Microprocessors & Assembly Language

Credit:03

Time: 1.00 Hour

Full Mark: 60. There are Four Questions. Answer three questions including Q-1 and Q-2.

1.
 - a. Define a microprocessor and individual functional units using a suitable block diagram. Why clock is essential in a microprocessor? How does it measure the processor performance? [10]
 - b. Define the individual elements and functions of Execution Unit (EU) in 8086 architectures. [10]
2.
 - a. Define segmentation. How may segments can be in a program? What are the purposes those serve? Maximum length of a segment is 64KB- Why? [10]
 - b. What is the difference between logical and physical address. Show the logical address format in 8086 and the procedure to calculate physical address from logical address. If the upper 16 bit of a starting address in a segment is 1240H, then find out the ending address of the segment. [10]
3.
 - a. Explain general and special roles for both AX and BX with suitable examples. [10]
 - b. Show the signed and unsigned range for 4-bit number system. [10]
Find out the status of six conditional flags after the execution of the following instruction with proper explanations.
ADD AL, BL
Where AL= (last two digits of your registration number) H and
BL= FFH

OR

4.
 - a. Write an assembly program to solve the following expression [10]
$$C = A^2 + 2AB + B^2$$

Here A, B and C are byte type memory variables, you can define those in program.
 - b. What is the difference between MOV and XCHG instruction? What will be the flag condition during execution of these instructions [10]

University of Asia Pacific

Department of Computer Science & Engineering

Mid-Semester Examination Fall -2021

Program: B. Sc. Engineering (3rd Year/ 1st Semester)

Course Title: Microprocessors and Assembly Language

Course No. CSE 311

Time: 1.00 Hours.

Credit: 3.00

Full Mark: 60

Answer three questions including Q-1 and Q-2.

1. a) Define microprocessor and briefly explain the tasks performed by it. [10]
Mention the major functions of BIU and EU in 8086 microprocessor.

- b) Evaluate the memory capacity or maximum addressable memory of a [10]
microprocessor containing the following specification:

Data Bus = 16 bit

Address Bus = 24 bit

Illustrate the corresponding memory architecture using bank layout.

2. a) 8086 is a 16 bit microprocessor but 20 bit addresses are too large to fit in [10]
16 bit registers. How can you resolve this issue?
Explain the process of generating physical address from logical address
in 8086 microprocessor.

- b) Suppose, you have 3 best friends named A, B, and C. [10]

CS = (Last 4 digit of your student id) H

IP = (Last 4 digit of your phone number) H

SS = (First 4 digit of your student id) H

SP = (First 4 digit of your phone number) H

Now, find out the physical address of the first instruction to be executed
and the first location of the Stack to be used to execute the code.

3. a) Perform the following operations and mention the values of each of these registers after you execute the operations assuming: [10]

AX = first 4 digits of your student ID

BX = last 4 digit of your student ID

CX = first 4 digit of your best friends student ID

DX = last 4 digits of your best friends student ID

i) MOV AX, DX

ii) INC CX

iii) ADD BX, AX

- b) Represent the flag bits in an 8086 microprocessor using appropriate figure. After the execution of the instructions stated in (Q-3a) show the status of the CF, PF, ZF, SF, OF Flags with proper justification. [10]

OR,

4. a) Let, AX = 4234H and BX = 1234H. [10]
Using these data briefly compare the following instructions:

i) MOV and XCHG

ii) CMP and SUB

iii) Register addressing mode and Immediate addressing mode

- b) Determine the addressing modes of the following instructions providing appropriate justification for your answer: [10]

i) XOR AX, DX

ii) ADD AL, 45

iii) MOV AX, [BX+DI]

iv) MOV AX, [SI]

v) SUB AL, [SI+BP+2000]

University of Asia Pacific
Department of Computer Science & Engineering
Mid-Semester Examination Spring 2021
Program: B. Sc. Engineering (3rd Year/ 1st Semester)

Course Title: Microprocessors & Assembly Language. Course No. CSE 311 Credit: 3.00

Time: 1.00 Hour.

Full Mark: 60

There are **Three** Questions. **Answer All.**

1. a. Define a microprocessor and discuss the importance of a microprocessor. If you design a microprocessor what will be characteristics you will incorporate? [10]
b. How can you relate the address and data bus to find out the capacity of memory? Evaluate the capacity of 8086 considering the size of address and data bus in 8086. [10]
2. a. Explain the special role of CX and DX registers. Define signed and unsigned numbers with appropriate scale. [10]
b. 'Queue operates in FIFO and Stack operates in LIFO' - explain their operations according to this phenomena. [10]
3. a. Write an assembly program to solve the following expression [10]
$$X = 2A - B$$

[Note: Consider and define three word type variables.]

b. Using the different functions of DOS Interrupt routine INT 21h do the followings: [10]
 - i. Display a prompt message "Welcome to CSE-UAP"
 - ii. Display the first character of your name.
 - iii. Input a character.

OR

- a. Consider three variables, A, B and C, where A= 08H, B= last two digits of your registration number in Hex and C= 1FH. Then write the active code to do the followings: [10]

- i. Add first two variables
 - ii. Swap last two variables
 - iii. Negative last variable
- b. For the above operations (Q-a) find out the status of six conditional Flags CF, PF, ZF, SF, AF and OF. **[10]**

University of Asia Pacific
Department of Computer Science & Engineering
Mid-Semester Examination Fall-2020
Program: B. Sc. Engineering (3rd Year/ 1st Semester)

Course Title: Microprocessors & Assembly Language. Course No. CSE 311 Credit: 3.00

Time: 1.00 Hour.

Full Mark: 60

There are **Four** Questions. **Answer three questions.**

1. a. What are the basic **components** of a microprocessor? Draw the block diagram of a whole computing system and mention each section. Why is **clock** mandatory in digital circuit? [10]
b. Write some lines by yourself about **pipelining concept** that has implemented elementary basis in 8086 architecture. How does the processor get speedup by this feature? [10]
2. a. Explain the role of **AX** and **DX** registers by **MUL**, **DIV** operations and **IN,OUT** operations, [10]
b. How does **SP** is used to operate stack by using **PUSH** and **POP** instructions, Write in your own words , if possible give an example. [10]
3. a. Write an assembly code to solve the following **expression** [10]
C=XA+YB
Here A, B and C are memory variables, you can define those in program. On the other hand X and Y are the constants.
Where
X= (last digit of your registration number MOD 2)
Y=(last digit of your registration number MOD 2)
[You can find out this part manually, example 5 MOD 2= 1]
b. What is the difference between **MOV** and **XCHG** Opcode? What is the role of Temp registers during XCHG operation? Give an example of sample code. [10]

OR

4. a. Define the six **conditional flags**, with the conditions those reflect. [10]

- b. ADD AX, BX. Where, AX = 8000H, BX = (Last four digits of your Phone number)H. After the execution of the instruction show the status of the flags CF, PF, ZF, SF, OF Flags. **[10]**

University of Asia Pacific
Department of Computer Science & Engineering
Mid-Semester Examination Spring-2020
Program: B. Sc. Engineering (3rd Year/ 1st Semester)

Course Title: Microprocessors & Assembly Language. Course No. CSE 311 Credit: 3.00
Time: 1.00 Hour. Full Mark: 60

There are **Four** Questions. **Answer three questions including Q-1 and Q-2.**

1. a. Suppose, you have 3 best friends named A, B, and C. [14]
CS = (Last 4 digit of your student id)H,
IP = (Last 4 digit of your best friend A's student id) H.
SS = (Last 4 digit of your best friend B's student id) H,
SP = (Last 4 digit of your best friend C's student id) H.
Now, find out the physical address of the first instruction to be executed and the first location of the Stack to be used to execute the code.
- b. With an example, prove that the logical address of a memory location is not unique. [6]
2. a. "A processor is N-bit" - what does it imply? From address bus, data bus and size of ALU, and registers, which effects the processor length? Shortly describe BIU. [10]
- b. Write a short note on the following (answer any 2). [10]
i. EU
ii. Stack
iii. ALU
iv. Status Register
3. a. ADD AX, BX. Where, AX = (Last 4 digit of your student id)H, BX = FFFF H. [10]
How the aforementioned instruction will affect CF, PF, ZF, SF, OF Flags.
- b. Using Only MOV, ADD, SUB, INC, DEC translate the following high-level language assignment statements into assembly language. A, B, and C are word variables. [10]
1. $C = \text{constant1} * A + \text{constant2} * B + \text{constant3}$
2. $C = 2 * (\text{constant1} * A + \text{constant2} * B + \text{constant3} * C)$
Where, $\text{constant1} = (\text{Student ID}) \bmod 3 + 1$
 $\text{constant2} = (\text{Student ID} + 1) \bmod 3 + 1$
 $\text{constant3} = (\text{Student ID} + 2) \bmod 3 + 1$

[Note: for Student ID = 12101022,
 $\text{Constant1} = (12101022) \bmod 3 + 1 = 0 + 1 = 1$

Constant2 = $(12101022+1) \bmod 3 + 1 = 1 + 1 = 2$
Constant1 = $(12101022+2) \bmod 3 + 1 = 2 + 1 = 3]$

OR

4. a. Compare based and indexed addressing modes. [6]
- b. Find out the addressing modes of the following instructions: [4]
- i. MOV AX, A[BP+SI]
 - ii. MOV AX, A[BX][DI]
 - iii. MOV AX, D
- c. Consider the following instructions and mention register used by these(answer any 4): [10]
- i. MUL
 - ii. OUT
 - iii. LOOP
 - iv. PUSH
 - v. INT 21H(for single character input)
 - vi. INT 21H(for single character output)
 - vii. INT 21H(for string output)

CSE 313
University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 2025
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Software Engineering
Time: 1 hour.

Credit Hour: 3.00

Course Code: CSE 313
Full Marks: 20

There are Two Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. **Compare** static, dynamic, and decentralized webs in the context of web-based software engineering. [5]
- b. **Explain** the risk management process of software engineering [5]

QUESTION 2 [10 MARKS]

A national space agency is developing the onboard software system for a deep-space spacecraft mission. The system will control critical functions such as trajectory corrections, communication with Earth, fuel management, and onboard diagnostics. Since the spacecraft will operate millions of kilometers away, any defect could result in the complete failure of the mission, which has taken years of planning and billions of dollars of investment.

Before development begins, scientists and engineers have already specified the requirements in great detail. These requirements must be strictly followed, as they are based on well-studied physical models, safety regulations, and international standards. Each stage of the project must be thoroughly documented and approved before moving forward. Due to the critical nature of the mission, the software will only be tested when the entire system is fully developed, and partial or trial versions would not be useful.

As the project manager:

- a. Propose the most suitable software process model for this project from the following options: RAD, Waterfall, Incremental, Agile, V-model. **Examine** your answer and list the steps of that process. [5]
- b. **Generate** five most important functional requirements of the spacecraft software system based on the scenario above. [5]

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Fall 24
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Software Engineering
Time: 1 hour.

Credit Hour: 3.00

Course Code: CSE 321
Full Marks: 20

There are Two Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Discuss the code of ethics in Software Engineering and examine the key issues related to professional responsibility in the field. [5]
- b. Compare Green Software Engineering and Sustainable Software Engineering, and determine which approach is more beneficial for nature. [5]

QUESTION 2 [10 MARKS]

A university plans to implement a Smart Campus Management System to streamline administrative tasks and enhance student services. The system should allow students to register for courses, check their grades, pay fees, and receive notifications about academic events. Faculty members should be able to manage course materials, upload grades, and track student progress. Additionally, administrators should have access to student records, financial reports, and faculty performance analytics. To ensure smooth implementation, the university has decided to follow an iterative and incremental development approach. However, before deployment, a new requirement has emerged: The system should now integrate an AI-powered recommendation engine to suggest courses based on a student's academic history and interests. This change must be accommodated in future increments without disrupting the already developed functionalities. [10]

Identify the functional and non-functional requirements of the Smart Campus Management System. Propose the most suitable software process model for this project. Justify why you chose this model over others by comparing its advantages and limitations with alternative models.

University of Asia Pacific

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Mid-Semester Examination

Spring-2024

3rd Year 2nd Semester

Course Code: CSE 321

Course Title: Software Engineering

Credit: 3.0

Time: 1.0 Hour.

Full Mark: 20

There are two questions. Answer all of them. Part marks are shown in the margins.

1. ☒ Compare static, dynamic, and decentralized webs in the context of web-based software engineering. [5] [CO1]

☒ Describe (names only) the most common key aspects of project planning elements in software project management. [5] [CO1]

☒ **User requirement of AI-chatbot :** [10] [CO2]

An AI chatbot is a web application or interface that is capable of engaging in human-like conversations with users. It is essential to provide customer support that is both scalable and efficient in order to assure client contentment and competitiveness in the modern business environment. An ideal solution, a chatbot that is facilitated by AI provides real-time support across multiple platforms and accumulates responses to common enquiries.

The goal of the undertaking is to develop an AI-driven chatbot that employs Natural Language Processing (NLP) to respond to intricate multilingual enquiries and swiftly solve common client questions. The chatbot must interface with the company's Customer Relationship Management (CRM) system to obtain consumer information, such as order status, and refer intricate enquiries to human agents. Over time, the chatbot should assimilate knowledge from exchanges, hence perpetually improving its response quality and accuracy. A new criterion has emerged before the application delivery. Assess customer satisfaction over the revised CRM integration and enhance the query escalation procedure.

Assume that you are a product manager of a reputed software company. Identify the most appropriate software process model for fulfilling the user's requirements. Correlate the functional requirements with the product backlogs of the project in accordance with the proposed software process model.

University of Asia Pacific
Department of Computer Science and Engineering
Program: B.Sc. in CSE

Mid-Semester Examination

Fall-2023

3rd Year 2nd Semester

Course Code: CSE 321

Course Title: Software Engineering

Credit: 3.0

Time: 1.0 Hour.

Full Mark: 20

There are two questions. Answer all of them. Part marks are shown in the margins.

1. a. Describe five advantages of decentralized applications (dApps). [5] [CO1]
b. Explain how data interoperability is achieved in Web 3.0 applications and its significance. [5] [CO1]
2. You are a product manager. You have a cross-functional team including developers, designers, testers, and product owners (product end-users). Now you're on your way to developing a cross-platform location-tracking application within 4 weeks. There is a challenge that 50% of the work has to be delivered to the customer within 2 weeks. [10] [CO2]

User Requirement:

Allow users to create accounts or log-in using existing credentials. Maintain user profiles and preferences. When developing the user authentication and profile setup functionalities, the product owner may suggest any changes to it. Continuously track and update the user's current location using GPS and display the user's real-time location on a map. Users may have the option to store location history and enable sharing options. Send geofencing and location-based notifications based on their current location and it will be a user-friendly notification system. Request appropriate permissions from the user to access their location data. Provide clear explanations for why location data is needed and how it will be used. Ensure that user location data is stored and transmitted securely. Comply with privacy regulations and protect user data from unauthorized access. Before delivery of the end product check overall application performance, optimize battery usage, and cross-platform stability, and gather feedback from users. At the last minute, the user suggests another requirement: the application must have offline map access capabilities. So, review all progress, address challenges and collect feedback from users for improvements before the product delivery deadline.

Identify the most appropriate software process model to fulfil the user requirements. Correlate all the project sprint and initial backlog of every sprint according to the proposed software process model.

University of Asia Pacific
Department of Computer Science and Engineering
Mid-Semester Examination Fall-2022

Program: B.Sc. in CSE

Course Title: Software Engineering
Time: 1.0 Hour.

Course No. CSE 321

Credit: 3.0

Full Mark: 60

There are **three (3)** questions. Answer **all** of them.

1.
 - a. What challenges might software engineers face in the Fourth Industrial Revolution (4IR)? [10]
 - b. How does AI technology lead to a more sustainable future? Give an example in the field of software engineering. [10]

Or

 - a. What challenges might software engineers face in the web 3.0? [10]
 - b. What are some of the most challenging aspects of changing a legacy system for a new one? Give an example in the field of software engineering. [10]
2.

Hybrid apps are mobile applications optimized for cross-platform compatibility and built with web technologies. A sustainable hybrid application meets customer requirements through early and continuous delivery.

During the development, there are higher chances of unknown risks that may expose and change the requirements even late in the project. The customer, developers, and application users should be able to maintain a constant pace indefinitely. Generally, the development of a mobile application is divided into sub-modules or small independent modules.

 - a. Which software process model is best suited to develop this type of application? Give an interpretation on behalf of your answer. [10]
 - b. How is DevOps different from Agile methodology in the case of hybrid application development? [10]
3. End-to-end encryption (E2EE) is a method of secure communication that prevents third parties from accessing data while it's transferred from one end system or device to another.

University of Asia Pacific
Department of Computer Science and Engineering
Mid-Semester Examination Fall-2021
Program: B.Sc. in CSE

Course Title: Software Engineering

Course No. CSE 321

Credit: 3.00

Time: 1.00 Hour.

Full Mark: 60

There are **Four** Questions. Answer three questions including Q-1 and Q-2.

1. ☒ Suppose your software company is assigned a project by a renowned Biology Lab for designing a software system for them, which will tract the RNA sequences of the Covid 19 virus and the changing symptoms of the patients. Then the system will predict the variant if given the symptoms of a patient. Which software process model will you use and why? Describe the model in accordance to your system. [20]
2. ☒ a. Draw the ER Diagram of the system mentioned in 1(a). [10]
☒ b. Do you think there is any privacy issue that you will face while building the systems in 1(a)? Explain in detail. [10]
3. ☒ a. What is System? Explain different types of system with examples. [10]
☒ b. What is Scrum? Explain the process flow of Scrum methodology with example. [10]

OR

4. ☒ a. What are the six types of risks that can rise? Explain each of them. [10]
☒ b. What is software design? What are the objectives of software design? [10]

rdd tdm

University of Asia Pacific
Department of Computer Science and Engineering
Mid-Semester Examination Spring-2021
Program: B.Sc. Engineering in CSE

Course Title: Software Engineering

Course No. CSE 321

Credit: 3.00

Time: 1.00 Hour.

Full Mark: 60

There are **Four** Questions. Answer three questions including **Q-3** and **Q-4**.

1. a. Mention the most appropriate generic software process model that might be used for managing the development of the following systems: [5]
 - I. Illumina: a simple DNA sequence analysis software, which may use SARS-CoV-2 genome sequencing.
- b. To help counter terrorism, many countries are planning the development of computer systems like video surveillance system that track large numbers of their citizens and their actions. Clearly, this has privacy implications for citizens. On other hand, tracing system helps to protect terrorism. [15]

Discuss the code of ethics based on software engineering of developing this type of system.
- Or**
2. a. Mention the most appropriate generic software process model that might be used for managing the development of the following systems: [5]
 - I. Proteus MMX: a cloud-based EAM software by Eagle CMMS designed to manage power plant.
- b. Spying software is a type of monitoring software that surreptitiously install on user's electrical device like cell phone, laptop, desktop, etc. Clearly, this has privacy implications for citizens. On other hand, monitoring system helps to protect suspicious movement of any citizens. [15]

Discuss the code of ethics based on software engineering of developing this type of system.
3. In software development, agile practices involve discovering requirements and developing solutions through the collaborative effort of self-organizing and cross-functional teams and their customers.
- a. Is agile the best methodology for mobile application development? Give an interpretation on behalf of your example. [10]
- b. Should you follow the agile methodology, When working on personal projects? [10]

4. Electronic payment (e-pay) system showed in figure is a subspecies of the payment system, which provides the process of passing online payment transactions over the Internet.

Simply put - payment systems are a way to pay for goods and services with a bankcard or electronic money (cashless). The most common payment systems in Bangladesh are debit card, master card and mobile banking. Merchant have a choice to select a payment system among the following payment system.

To make a payment through e-pay consumer may need to go authentication process. Payment gateway will ensure the user authentication security. Corresponding financial authorities will responsible for payment gateway. Finally, Bank will pay money to the merchant and complete the payment transaction.

- a. List all the functional, enduring and volatile requirements for this e-pay system. [10]
- b. Draw a block diagram for e-pay system according to your listed requirements. [10]

University of Asia Pacific
Department of Computer Science and Engineering
Mid-Semester Examination Fall-2020
Program: B.Sc. Engineering

Course Title: Software Engineering

Course No. CSE 321

Credit: 3.00

Time: 1.00 Hour.

Full Mark: 60

There are **Four** Questions. Answer three questions including **Q-3** and **Q-4**.

1. a. Giving reasons for your answer based on the type of system being developed, suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following systems: [5]
 1. A virtual reality system to support software maintenance
 - b. To help counter terrorism, many countries are planning the development of computer systems that track large numbers of their citizens and their actions. Clearly, this has privacy implications for citizens. On other hand, tracing system helps to protect terrorism. Discuss the code of ethics based on professional aspect according to software engineering of developing this type of system. [15]
- Or**
2. a. Giving reasons for your answer based on the type of system being developed, suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following systems: [5]
 1. A university accounting system that replaces an existing system
 - b. Spying software is a type of monitoring software that surreptitiously install on user's electrical device like cell phone, laptop, desktop, etc. Clearly, this has privacy implications for citizens. On other hand, monitoring system helps to protect suspicious movement of any citizens. Discuss the code of ethics based on public-welfare according to software engineering of developing this type of system. [15]
3. In software development, agile practices involve discovering requirements and developing solutions through the collaborative effort of self-organizing and cross-functional teams and their customers.
 - a. Write an example that use the agile model for development. Give an interpretation on behalf of your example. [10]
 - b. How does extreme programming methodology assist to elaborate your proposed example? [10]

4. Electronic payment (e-pay) system showed in figure is a subspecies of the payment system, which provides the process of passing online payment transactions over the Internet.

Simply put - payment systems are a way to pay for goods and services with a bankcard or electronic money (cashless). The most common payment systems in Bangladesh are debit card, master card and mobile banking. To make payment through e-pay consumer may need to go authentication process. Payment gateway will ensure the user authentication security. Corresponding financial authorities will responsible for payment gateway. Finally, Bank will pay money to the merchant and complete the payment transaction.

- a. List enduring and volatile requirements for the e-pay system. [5]
b. Draw the use case diagram for e-pay system. [15]



Fig: Electronic payment (e-pay) system

University of Asia Pacific
Department of Computer Science and Engineering
Mid-Semester Examination Spring-2020
Program: B.Sc. Engineering

Course Title: Software Engineering

Course No.: CSE 321

Credit: 3.00

Time: 1.00 Hour.

Full Mark: 60

There are **Four** Questions. **Answer three questions including Q-1 and Q-2.**

1. a. What are the problems and challenges that software engineers are likely to face in the 21st century? How to overcome those challenges? [10]

b. To help counter terrorism, many countries are planning the development of computer systems that track large numbers of their citizens and their actions. Clearly this has privacy implications for citizens. On other hand, tracing system helps to protect terrorism.
Discuss the ethics according to software engineering of developing this type of system. [10]
2. a. Draw a use case diagram for Online Examination System. The system includes the following information: [10]

The University of Asia Pacific (administrator) has a facility to conduct an online exam by providing questions to the students and the administrator to verify the details of the students are correct or not. Then, the administrator provides authentication to write the exam of the verified students. After conducting the exam the administrator needs to pay attention to manage the examination. Finally, the admin gets the result details.

Every student has permission to login the system, before that the student needs to complete the registration process. After that student needs to select the required exam from the list and gets the selected question paper from the administrator. After answering each question the student submits the answer script of the predefined procedure and gets the result instantly.

b. According to question no. 2 (a) to register Online Examination System, every student needs to fill up following information: Name, Roll, Gender, Department, university provided E-mail address and Password. [10]

Sudip

Create an object of registration class using your personal information according to UML object diagram.

3. a. Giving reasons for your answer based on the type of system being developed, suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following systems: [10]
 - A virtual reality system to support software maintenance
- b. What are the industry based examples of Prototyping Model? Give an explanation on behalf of your example. [10]

OR

4. a. Giving reasons for your answer based on the type of system being developed, suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following systems: [10]
 - A university accounting system that replaces an existing system
- b. What are the industry based examples of Agile Model? Give an explanation on behalf of your example. [10]

CSE317

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Fall 2025
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Computer and Cyber Security
Time: 1 hour.

Credit Hour: 3.00

Course Code: CSE 317
Full Marks: 20

There are Two Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Explain the ten security principles of cyber security. [5]
- b. Apply shift cipher to encrypt the given message where, [5]
 $K = (\text{last 2 digits of your student ID} \% 10) + 10$
Message = "UAP IS COMMITTED TO EXCELLENCE"

QUESTION 2 [10 MARKS]

- a. Describe the steps of the AES algorithm using the flowchart. [5]
- b. Sketch the Diffie-Hellman key exchange mechanism and find the shared secret key [5]
between Alice and Bob using the following data.

Alice

secret_key, $a=7$

public_key
{ $g=3, p=17$ }

Bob

secret_key, $b=11$

PHY 301

University of the Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 2025
Program: B.Sc. in CSE
3rd Year 2nd Semester

Course Title: Modern and Quantum Physics
Time: 1 hour

Credit Hour: 3.00

Course Code: PHY 301
Full Marks: 20

There are Two Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- State Planck's quantum hypothesis and explain blackbody radiation in terms of it. [5.0]
- Show that the energy of an electron for an atom in n^{th} orbit [3.0]

$$E_n = -\frac{me^4Z^2}{8\epsilon_0^2n^2h^2}$$

where the symbols have their usual meanings.

- Show that for hydrogen atom the total energy of an electron in ground state is -13.6 eV . [2.0]

QUESTION 2 [10 MARKS]

- Describe photoelectric effect. Mention the experimental observations of photoelectric effect. [3.0]
- Explain Einstein's photoelectric effect equation. [2.0]
- A photon of wavelength 400 nm strikes a metal surface with a work function of 2.2 eV. Calculate the kinetic energy of the emitted electron. (Planck's constant $h = 6.63 \times 10^{-34} \text{ Js}$; Speed of light $c = 3.0 \times 10^8 \text{ m/s}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$) [5.0]